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SUPERSEDING
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MILITARY SPECIFICATION

THERMOMETER, INDICATING, CAPILLARY TUBE AND BULB: FLUSH MOUNTING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of indicating dial thermometer with a temperature range of minus 20 Fahrenheit (F.) to plus 120 F. (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-A-548 - Antifreeze, Ethylene Glycol, Inhibited
PPP-B-636 - Box, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MS 27211 - Thermometer, Indicating, Capillary Tube and Bulb

(Copies of specifications and standards required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6685

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread-Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402).

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D.C. 20036).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified (see 6.2), before production is commenced, a sample thermometer covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of the responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The thermometers delivered under this specification shall be the manufacturer's commercially current standard product, except for any changes necessary to comply with specification requirements.

3.3 Materials.- Materials shall conform to the applicable specifications and requirements specified herein and on MS 27211. Materials not definitely specified shall be of the quality normally used for the purpose. Commercial tolerances shall apply to material thickness and diameters.

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3.4 Design and construction.- The design and construction of the thermometers shall conform to the requirements specified herein and on MS 27211. Thermometers shall be flush mounting with a temperature range of minus 20F. to plus 120F. The thermometer shall operate on a vapor tension principle through an actuating medium consisting of a gear mechanism, bourdon tube, capillary tube and a sensing element or bulb. The indicating pointer shall be connected to the bourdon tube in such a manner to indicate on a circular design dial graduated in the Fahrenheit system.

3.4.1 Indicating mechanism.- The indicating mechanism shall operate under the vapor pressure of a halogen chemical fluid acting within a bourdon tube. The indicating mechanism shall be the conventional "C" configuration or spiral pattern, both having the anchor end in a socket and the free end connected to the indicating pointer through a single lever, magnet or direct coupling. The pointer shall rotate in a clockwise manner on a temperature increase and in reverse on temperature decrease. The mechanism shall have a spindle pinion and geared sector. If a conventional "C" tube is used, the pointer dampening of the spindle action shall be by means of a flat spiral wound spring. If the construction is of the pinion-and-sector design, the sector shall be provided with milled teeth and shall be not less than 0.0478 inch thick. The connecting links shall be not less than 0.0478 inch thick. Pinion, sector, and linkage shall be made of corrosion-resistant steel.

3.4.2 Calibration adjustment.- A calibration adjustment shall be provided to allow temperature reading correction of the instrument.

3.4.3 Brazing.- The sensing bulb, capillary tubing socket, and bourdon tube shall be hermetically sealed by brazing. Tightness of brazed components shall be determined in accordance with the test specified in 4.8.1.

3.4.4 Performance.- The thermometer shall be accurate to within plus or minus 3 F. at minus 20 F., 1-1/2 F. at midpoint of temperature range and 1 F. at 120 F. The ascending and descending temperature response of the entire temperature range shall be of a rate of not more than 30 seconds maximum time lag when tested as specified in 4.8.2.

3.4.5 Indicating dial.- The face of the dial shall be 2-1/2 inches in diameter and shall have a white background with black Arabic numerals on the inside of a graduated radial scale sector comprising an arc of approximately 270 degrees and with a temperature range of from minus 20 F. to plus 120 F. The word "FAHRENHEIT" in black and of legible size shall be on the face of the dial.

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3.4.6 Crystal.- The crystal shall be clear and free from blemishes. The crystal shall be secured to the bezel by (1) a gasketed screw ring; (2) a gasket and split snapring; or (3) a gasket in back of the crystal that fits into a groove in the bezel.

3.4.7 Bezel.- The bezel shall be either the slip-one type, fastened to the case with machine screws, or shall be threaded with matching threads on the case. Both methods of fastening shall be made with rim gaskets.

3.4.8 Screws, machine.- The machines screws shall be brass. The screw threads shall conform to the National Bureau of Standards Handbook H-28.

3.5 Finish.- The finish shall be in accordance with the manufacturer's commercial practice and shall not contain any burrs, slivers, or sharp edges.

3.6 Marking for identification.- The manufacturer's name or trademark and model number shall be marked on the dial face approximately 1/4 inch from bottom edge.

3.7 Workmanship.- All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. Shearing and chipping shall be done accurately and cleanly on all portions of the work and shall be neatly finished. All bolt holes shall be accurately and clearly punched or drilled. Bolts, nuts, and screws, when used, shall be drawn tight and so secured that they will not work loose.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Preproduction sample inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I, dimensions as specified, and tested as specified in 4.8.1 and 4.8.2.

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* 4.3 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, modified, or qualified in this specification or applicable purchase document.

* 4.4 Sampling for inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

* 4.5 End item inspection.- The lot shall be the number of thermometers offered for inspection at one time. The sample unit shall be one complete thermometer.

* 4.5.1 Visual examination.- The thermometer shall be examined for defects in table I. The inspection level shall be II, and the acceptable quality level (AQL) shall be 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. - Visual classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as specified		X
	Contains burrs, slivers, or sharp edges	X	
Design, construction and workmanship	Scale not graduated in Fahrenheit system on a scale of circular design	X	
	Bolt holes not accurately drilled or punched		X
	Bolts, nuts or screws missing or loose		X
	Shearing not cleanly and accurately done		X
	Temperature range not minus 20 F. to plus 120 F.	X	
	Thermometers not flush mounting	X	
	Machine screws not brass		X
	Actuating and indicating mechanism not as specified	X	

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TABLE I. - Visual classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Calibration adjustment	Missing	X	
Indicating dial	Not a white background with black Arabic numerals	X	
	The word Fahrenheit missing, illegible or not black		X
Crystal	Not clear and free from blemishes		X
	Not secured by a gasketed screw ring, a gasket and split snapring, or a gasket in back of crystal that fits into groove of bezel		X
Bezel	Not slip on type, fastened with machine screws or threaded with matching threads on case		X
	Rim gaskets missing	X	
Marking for identification	Missing, incomplete, illegible, not marked on dial face or not marked in prescribed manner	X	

* 4.5.2 Dimensional examination.- Examination shall be made of the completed thermometers to determine compliance with this specification and MS 27211. Any dimension not within the specified requirement, shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.6 End item testing.- Thermometers shall be tested in accordance with 4.8.1 and 4.8.2. The inspection level shall be S-2 with an AQL of 1.5, expressed in terms of defects per hundred units.

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4.7 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as set forth in table II. The lot shall be the number of containers offered for inspection at one time. The sample unit shall be one container fully prepared for delivery, except the container need not be sealed. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units. Examination for closure defects shall be made of shipping containers fully prepared for delivery.

TABLE II. - Examination of preparation for delivery

Examine	Defect
Marking, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Material	Component missing, or damaged
Content	Number per container is more or less than required

4.7.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.8 Tests.-

4.8.1 Leakage test.- Place the assembled thermometer in an enclosed leakproof container for a period of 24 hours and at an ambient temperature of 75 degrees F., plus or minus 5 degrees F. Upon opening the container, check contents with a halogen leak detector capable of detecting leaks of 1/20 of an ounce per year to determine compliance with 3.4.3. Any indication of leakage shall constitute a failure.

4.8.2 Performance test.- The thermometer shall be tested for performance with the dial and capillary tubing in a room having an ambient temperature of 75 degrees F., plus or minus 5 degrees F., and the sensing bulb inserted in a solution consisting of 45 percent by volume of diethylene-glycol specified in O-A-548 and water. The

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sensing bulb shall be subject to the temperature range specified in 3.4.4. Calibration curves of time lag shall be recorded and evaluated to determine compliance with the requirements specified in 3.4.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each thermometer shall be packaged in a snug-fitting box conforming to style TS, type CF, variety optional, class domestic, grade 200 of PPP-B-636. The contents shall be immobilized with cushioning or scored and die cut fiberboard sheets made from the same material as the box. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C.- Thermometers shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Twelve thermometers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- Unless otherwise specified (see 5.2.2.1), twelve thermometers packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), shipping containers shall be fabricated and closed in accordance with V3c or V3s of PPP-B-636.

5.2.3 Level C.- Thermometers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- The thermometer covered by this specification is intended to be used for indicating temperatures of an inclosed refrigerated air space, air duct for air conditioning installations, or an air stream entering or leaving an evaporator coil. This thermometer is not recommended for temperatures below freezing.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When preproduction sample is waived (see 3.1).
(c) When weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 6685-0168

Review activities:

Army - MD
Navy - YD
Air Force - 82

User activities:

Army - CE
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Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-9955A

Thermometer, Indicating, Capillary Tube and Bulb: Flush Mounting

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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